DBMS ASSIGNMENT

ASSIGNMENT 7

Sample Data Insertion

Project: Library Loan Management System

Database: MySQL 8.0

Objective

Insert realistic sample data into all six library tables and verify the record counts and borrowing states.

Note: outputs are expected results based on the included sample data. Run the SQL in your MySQL environment and insert your own screenshots if required by your instructor.

1. Dataset and Insertion Order

The sample dataset contains 10 members, 8 books, 9 authors, 10 book-author links, 16 physical copies, and 20 loans (73 rows in total). Sample names and email addresses are fictional. Insert parent records before child records and run the script once in a fresh database.

2. SQL INSERT Statements

```
USE LibraryLoanDB;
START TRANSACTION;

INSERT INTO Members
    (MemberID, FullName, Email, Phone, JoinedOn, MembershipStatus) VALUES
(21, 'Asha', 'asha@example.com', NULL, '2026-08-01', 'Active'),
(22, 'Ravi', 'ravi@example.com', NULL, '2026-08-03', 'Active'),
(23, 'Meera', 'meera@example.com', NULL, '2026-08-05', 'Inactive'),
(24, 'Neha', 'neha@example.com', NULL, '2026-08-08', 'Active'),
(25, 'Arjun', 'arjun@example.com', NULL, '2026-08-10', 'Active'),
(26, 'Priya', 'priya@example.com', NULL, '2026-08-12', 'Active'),
(27, 'Kabir', 'kabir@example.com', NULL, '2026-08-15', 'Active'),
(28, 'Ishita', 'ishita@example.com', NULL, '2026-09-01', 'Active'),
(29, 'Dev', 'dev@example.com', NULL, '2026-09-05', 'Active'),
(30, 'Sana', 'sana@example.com', NULL, '2026-10-01', 'Active');

INSERT INTO Books
    (BookID, ISBN, Title, Edition, PublicationYear, Category) VALUES
(101, NULL, 'Pride and Prejudice', NULL, 1813, 'Fiction'),
(102, NULL, 'Database Practice Workbook', '1st', 2026, 'Education'),
(103, NULL, 'Python Lab Notes', '1st', 2025, 'Computing'),
(104, NULL, 'Networks Made Simple', '2nd', 2024, 'Computing'),
(105, NULL, 'The Rainy Platform', '1st', 2023, 'Fiction'),
(106, NULL, 'Everyday Astronomy', '1st', 2022, 'Science'),
(107, NULL, 'Statistics Through Examples', '2nd', 2025, 'Education'),
(108, NULL, 'Campus Garden Handbook', '1st', 2026, 'Science');

INSERT INTO Authors (AuthorID, AuthorName) VALUES
(12, 'Jane Austen'), (13, 'Arun Rao'), (14, 'Nisha Das'),
(15, 'Kiran Shah'), (16, 'Leela Menon'), (17, 'Rohan Sen'),
(18, 'Farah Ali'), (19, 'Vivek Nair'), (20, 'Anita Bose');

INSERT INTO BookAuthors (BookID, AuthorID) VALUES
(101,12), (102,13), (102,14), (103,15), (104,15),
(104,16), (105,17), (106,18), (107,19), (108,20);

INSERT INTO BookCopies
    (CopyID, BookID, AccessionNumber, ShelfLocation, ConditionStatus) VALUES
(501,101,'LIB-001','A2-S3','Good'),
(502,101,'LIB-002','A2-S3','Good'),
(503,102,'LIB-003','B1-S1','Good'),
(504,102,'LIB-004','B1-S1','Damaged'),
(505,102,'LIB-005','B1-S1','Good'),
(506,103,'LIB-006','B2-S1','Good'),
(507,103,'LIB-007','B2-S1','Good'),
(508,104,'LIB-008','B2-S2','Good'),
(509,104,'LIB-009','B2-S2','Lost'),
(510,105,'LIB-010','A2-S4','Good'),
(511,105,'LIB-011','A2-S4','Good'),
(512,106,'LIB-012','C1-S1','Good'),
(513,106,'LIB-013','C1-S1','Good'),
(514,107,'LIB-014','B1-S2','Good'),
(515,107,'LIB-015','B1-S2','Withdrawn'),
(516,108,'LIB-016','C1-S2','Good');

INSERT INTO Loans
    (LoanID, MemberID, CopyID, BorrowedOn, DueOn, ReturnedOn) VALUES
(9000,22,501,'2026-08-10','2026-08-24','2026-08-22'),
(9001,21,501,'2026-09-07','2026-09-21',NULL),
(9002,22,503,'2026-08-20','2026-09-03',NULL),
(9003,21,505,'2026-08-15','2026-08-29','2026-08-27'),
(9004,24,506,'2026-08-12','2026-08-26','2026-08-24'),
(9005,25,508,'2026-08-15','2026-08-29','2026-09-02'),
(9006,26,510,'2026-08-18','2026-09-01','2026-08-30'),
(9007,27,512,'2026-08-20','2026-09-03','2026-09-05'),
(9008,24,514,'2026-09-01','2026-09-15','2026-09-13'),
(9009,28,502,'2026-09-02','2026-09-16','2026-09-15'),
(9010,25,506,'2026-09-05','2026-09-19','2026-09-22'),
(9011,29,507,'2026-09-10','2026-09-24','2026-09-20'),
(9012,26,511,'2026-09-12','2026-09-26','2026-09-26'),
(9013,27,513,'2026-09-15','2026-09-29','2026-10-01'),
(9014,28,505,'2026-10-01','2026-10-15','2026-10-10'),
(9015,24,506,'2026-10-05','2026-10-19',NULL),
(9016,25,508,'2026-10-10','2026-10-24',NULL),
(9017,26,510,'2026-10-12','2026-10-26',NULL),
(9018,27,512,'2026-10-15','2026-10-29',NULL),
(9019,29,514,'2026-10-18','2026-11-01',NULL);

COMMIT;
```

3. Verification After Insertion

```
SELECT 'Members' AS TableName, COUNT(*) AS Records FROM Members
UNION ALL SELECT 'Books', COUNT(*) FROM Books
UNION ALL SELECT 'Authors', COUNT(*) FROM Authors
UNION ALL SELECT 'BookAuthors', COUNT(*) FROM BookAuthors
UNION ALL SELECT 'BookCopies', COUNT(*) FROM BookCopies
UNION ALL SELECT 'Loans', COUNT(*) FROM Loans;

SELECT COUNT(*) AS TotalLoans,
       SUM(ReturnedOn IS NULL) AS OpenLoans,
       SUM(ReturnedOn IS NOT NULL) AS ReturnedLoans
FROM Loans;

SELECT LoanID, CopyID, ReturnedOn, OpenCopyID
FROM Loans WHERE LoanID IN (9000, 9001) ORDER BY LoanID;
```

4. Expected Output

Table counts: Members 10; Books 8; Authors 9; BookAuthors 10; BookCopies 16; Loans 20. Total rows across all six tables = 73.

Loan summary: TotalLoans = 20, OpenLoans = 7, ReturnedLoans = 13.

Loans 9000 and 9001 both refer to copy 501 at different times. Loan 9000 has a return date and OpenCopyID = NULL. Loan 9001 is open and OpenCopyID = 501.

5. Cases Covered

The sample includes an inactive member, an active member with no loans, co-authored titles, a book never borrowed, repeated borrowing of the same copy at different times, and copies marked Damaged, Lost, and Withdrawn. It includes early, on-time, and late returns. Each borrowing date is on or after the member's join date.

6. Rerun Guidance

If you rerun the inserts, duplicate primary keys and unique values will cause errors. Use a fresh test database, or delete child rows before parent rows in the order Loans, BookCopies, BookAuthors, Authors, Books, Members. Do not delete real library records to repeat a lab.

7. Conclusion

The sample records populate all six tables and support the query reports in Assignments 8 and 9.